NAAN MUDHALVAN – PROJECT REPORT

SCRIPT-CONTROLLED ACL – RESTRICT RECORD ACCESS BASED ON FIELD VALUE

A Project Report Submitted as Part of the Naan Mudhalvan Training

Submitted by

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CERTIFICATE

This is to certify that the project report entitled “Script-Controlled ACL – Restrict Record Access Based on Field Value” is a bonafide work carried out by the students listed below during the academic year 2026–2027 as part of the Naan Mudhalvan training/project work.

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The project focuses on implementing a Script-Controlled Access Control List (ACL) in ServiceNow to restrict record access based on a specified field value. The project demonstrates record-level security, conditional access, script-based authorization, and testing of permitted and restricted access.

Faculty / Trainer Signature

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DECLARATION

We hereby declare that the project report entitled “Script-Controlled ACL – Restrict Record Access Based on Field Value” is prepared as part of the Naan Mudhalvan training/project activity. The report presents the concepts, requirement analysis, security design, implementation approach, testing procedure, and expected results of a Script-Controlled ACL in ServiceNow.

The project demonstrates how record access can be controlled using a field value and an ACL script. Wherever external technical documentation has been consulted, it is acknowledged in the References section.

Place: _____

Date: _____

Signature: _____

ACKNOWLEDGEMENT

We sincerely thank our college management and faculty members for providing the opportunity to participate in the Naan Mudhalvan training programme. We are grateful to the trainers and mentors who provided guidance on ServiceNow, Access Control Lists, record security, scripting, and IT Service Management.

We also thank our department for providing the required academic support to complete this project report. Finally, we thank everyone who supported us directly or indirectly during the preparation, testing, and documentation of this project.

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1. BRAINSTORMING & IDEATION PHASE

1.1 Project Overview

The project “Script-Controlled ACL – Restrict Record Access Based on Field Value” is designed around record-level security in ServiceNow. The main idea is to ensure that users can access records only when a defined field value satisfies the security condition.

1.2 Problem Identification

Some ServiceNow records may contain information that should only be accessible to authorized users or users associated with a particular field value. Without suitable access controls, users may access records outside the intended scope.

1.3 Proposed Idea

Configure an Access Control List and use a script to evaluate a field value before granting access. When the condition is satisfied, the ACL allows the requested operation; otherwise, access is restricted.

1.4 Objectives

• Understand ServiceNow Access Control Lists.

• Implement a record-level ACL for controlled access.

• Use a field value as part of the access condition.

• Apply a script-based authorization check.

• Restrict unauthorized record access.

• Test both allowed and denied access conditions.

1.5 Expected Outcome

The expected outcome is a secured ServiceNow record where access is granted only when the configured field-based condition is satisfied.

2. REQUIREMENT ANALYSIS PHASE

2.1 Functional Requirements

• The system shall identify the requested record and its relevant field value.

• An ACL shall control access to the protected record.

• The ACL shall evaluate the configured field-based condition.

• Authorized users shall be permitted to perform the configured operation.

• Unauthorized users shall be denied access.

• The access rule shall be tested with matching and non-matching field values.

2.2 Non-Functional Requirements

The access rule should be understandable, maintainable, secure, and consistent. The script should contain only the logic required for the authorization decision.

2.3 User Requirements

Authorized users need access to relevant records, while unauthorized users should not be able to access protected records.

2.4 Technical Requirements

• ServiceNow instance with appropriate access.

• A table containing protected records.

• A field whose value is used for the access decision.

• ACL configuration and scripting capability.

• Browser access.

• Controlled test users or conditions.

2.5 Requirement Traceability

The protected record requirement is connected to the ACL configuration. The field-based security condition is implemented through the ACL script, while testing verifies permitted and restricted conditions.

3. PROJECT DESIGN PHASE

3.1 Design Approach

The design separates the security requirement into the protected table/record, ACL operation, field-value condition, script evaluation, and access result.

3.2 Record Security Design

The target record contains a field whose value determines whether access should be allowed. The ACL is applied to the required operation according to the project requirement.

3.3 ACL Script Design

The ACL script evaluates the relevant record field and sets the authorization result accordingly. When the defined condition is satisfied, access is allowed; otherwise it is restricted.

3.4 Logical Flow

User requests record access → ServiceNow evaluates the ACL → ACL script checks the configured field value → condition is evaluated → access is allowed when satisfied → otherwise access is restricted.

3.5 Design Principles

• Protect only the intended table, record, and operation.

• Use meaningful names for ACLs and scripts.

• Keep the authorization condition simple and clear.

• Avoid unnecessary scripting.

• Test both allowed and denied conditions.

4. PROJECT PLANNING PHASE

4.1 Work Breakdown

• Understand the record access problem.

• List functional, non-functional, user, and technical requirements.

• Design the ACL, field condition, script, and access decision.

• Prepare implementation and testing sequence.

• Configure and implement the ACL and script.

• Execute allowed and restricted access tests.

• Prepare report, screenshots, configuration details, and references.

• Demonstrate the completed access-control behaviour.

4.2 Implementation Schedule

The implementation is planned in small steps so the ACL can be tested before final demonstration. The table, operation, field, ACL, and script are configured and then tested with different field values.

4.3 Risk Considerations

Possible issues include wrong ACL operation, incorrect field reference, script errors, conflicting ACL rules, and testing only one access condition. These risks are reduced through clear configuration and controlled testing.

5. PROJECT DEVELOPMENT PHASE

5.1 Development Environment

The project is implemented in ServiceNow using Access Control Lists. The target table contains the records to be protected, while the selected field provides the value used in the access decision.

5.2 ACL Configuration

An ACL is configured for the required table and operation. The ACL controls access to selected records according to the security requirement.

5.3 Script-Controlled Access

The ACL script evaluates the target field value. If the configured field condition is satisfied, the authorization result permits the operation. Otherwise, the operation is restricted.

5.4 Implementation Sequence

• Open the ServiceNow ACL configuration area.

• Identify the table and required operation.

• Create the appropriate ACL.

• Identify the field used for the access decision.

• Configure the ACL script.

• Save the ACL configuration.

• Test matching and non-matching field values.

• Verify that the access result matches the requirement.

5.5 Implementation Result

The completed configuration provides field-based record access control using a Script-Controlled ACL and demonstrates how ServiceNow can evaluate record information before allowing or restricting an operation.

6. PROJECT TESTING PHASE

6.1 Testing Objective

Testing verifies that the ACL script produces the expected access decision for different field values and user conditions. Both permitted and restricted scenarios are included.

6.2 Test Cases

Test Case	Expected Result	Status
TC01 – Open protected record	ACL is evaluated successfully.	Pass/To be recorded
TC02 – Matching field value	Access is allowed according to the rule.	Pass/To be recorded
TC03 – Non-matching field value	Access is restricted according to the rule.	Pass/To be recorded
TC04 – Unauthorized user	Protected operation is denied.	Pass/To be recorded
TC05 – Authorized condition	Configured operation is permitted.	Pass/To be recorded
TC06 – Repeated access	ACL behaviour remains consistent.	Pass/To be recorded

6.3 Testing Result

Testing should be completed in the actual ServiceNow instance before final submission. The final report should use the real observed results and screenshots from the completed ACL configuration.

6.4 Defect Handling

If access does not behave as expected, check the table, ACL operation, field reference, script condition, user roles, and other applicable ACL rules. After correction, execute the same test again.

7. PROJECT DOCUMENTATION PHASE

7.1 Purpose of Documentation

Documentation records the project objective, requirements, security design, ACL implementation, testing, and final access behaviour.

7.2 ACL Documentation

The ACL should be documented with its name, table, operation, field condition, script behaviour, and expected access result.

7.3 ACL Documentation Format

• ACL Name: _____

• Table: _____

• Operation: Read / Write / Create / Delete

• Field Used: _____

• Condition/Purpose: _____

• Script Functionality: _____

• Expected Result: _____

7.4 Screenshot Documentation

Suitable screenshots include the ACL configuration, script/condition, protected record, allowed access result, denied access result, and final successful behaviour.

7.5 References

ServiceNow learning material and documentation used during the Naan Mudhalvan ServiceNow training should be listed here. Only resources actually consulted should be included.

8. PROJECT DEMONSTRATION PHASE

8.1 Demonstration Objective

The demonstration presents the completed record-access restriction from start to finish, including the protected record, ACL configuration, field value, script-based decision, and access result.

8.2 Demonstration Flow

• Introduce the project title.

• Open the protected ServiceNow record.

• Show the relevant ACL configuration.

• Show the field used for the access decision.

• Demonstrate access when the field value satisfies the condition.

• Demonstrate a non-matching field value.

• Show that the protected operation is restricted.

• Summarize the final security result.

8.3 Demonstration Evidence

The final evidence should be based on the actual project instance. Screenshots should show the implemented ACL and its visible access-control result.

8.4 Final Outcome

The project demonstrates practical understanding of ServiceNow record-level security using a Script-Controlled ACL and field-based authorization.

CONCLUSION

The “Script-Controlled ACL – Restrict Record Access Based on Field Value” project provides a practical implementation of record-level security in ServiceNow. The project covers brainstorming, requirements, design, planning, development, testing, documentation, and demonstration.

The implementation focuses on ACL configuration, field-based access conditions, script-controlled authorization, and testing of allowed and restricted access. Actual ServiceNow screenshots and observed test results should be inserted after implementation is completed in the ServiceNow instance.

REFERENCES

• ServiceNow learning resources provided as part of Naan Mudhalvan ServiceNow training.

• ServiceNow platform documentation consulted during implementation.

• Project-specific training/reference materials provided during project development sessions.